

AXI4 Multi-Descriptor DMA Subsystem

Functional Verification Report

1. Executive Summary

This report summarizes the IP-level functional verification of a descriptor-based AXI4 memory-to-memory DMA subsystem. The design integrates an AXI-Lite control plane, a descriptor manager with request and completion FIFOs, vendor AXI DMA read/write engines, an AXI-Stream FIFO, and an AXI4 memory interface.

The verification environment was developed using SystemVerilog and UVM methodology with RAL frontdoor access, constrained-random and directed sequences, AXI-Lite/AXI monitors, a reference memory model, transaction-level scoreboard checking, functional coverage, SVA protocol checks, and Synopsys VCS/URG regression infrastructure.

Verification Status: PASS - Functional verification closure achieved.

Item	Result
Total UVM tests	28
Methodology	SystemVerilog + UVM + UVM RAL
Simulator	Synopsys VCS V-2023.12-SP2
Design focus	AXI4 multi-descriptor memory-to-memory DMA
Functional coverage	100% group coverage; reviewed bins closed

2. Verification Environment

The environment contains an AXI-Lite master agent, AXI memory-side monitoring/modeling, DMA descriptor and completion monitors, a RAL model and adapter, a reference memory model, a scoreboard, functional coverage, SVA interfaces, and regression/test selection infrastructure.

Component	Responsibility
AXI-Lite agent + RAL	Drive frontdoor register accesses, WSTRB combinations, commands, and status reads.
Descriptor manager monitor	Observe descriptor acceptance/rejection, queue level, active task, and completion records.
AXI memory model/proxy	Model AXI read/write responses, latency, backpressure, and SLVERR/DECERR injection.
Reference memory model	Maintain expected byte-addressable source and destination contents.
Scoreboard	Compare source/destination memory effects and completion metadata; report UVM errors on mismatch.
Coverage collector	Sample AXI-Lite, descriptor, FIFO, boundary, error, IRQ, reset, and outstanding points.
SVA checker	Check stability and valid/ready behavior during AXI-Lite, descriptor, AXI, and AXIS stalls.

3. Regression and Scenario Coverage

The final regression is organized into four scenario groups. The sequence library combines normal traffic with targeted boundary, protection, error, queue, reset, and protocol-stall scenarios.

Scenario group	Representative tests	Verification intent
Basic and RAL access	random_smoke, ral_smoke, wstrb, command_noop, RO protection, access policy	Control-plane programming, register semantics, and protection behavior.
Descriptor validity and queue	invalid descriptor, length sweep, alignment/range/overlap rejection, queue mid/full	Admission checks, FIFO capacity, rejection paths, and sticky status.
AXI transfer, error and backpressure	4KB matrix, sub-beat/TAG, read/write SLVERR/DECERR, backpressure, outstanding, write corner, toggle stress	Burst splitting, response errors, latency, stalls, FIFO propagation, and write-engine corners.
Completion, IRQ and reset	IRQ modes, empty completion, reset recovery, SVA activation	Completion lifecycle, level IRQ semantics, reset cleanup, and assertion activation.

4. Functional Coverage Closure

Coverage closure was performed with merged VCS/URG data. Functional bins were reviewed against the RTL feature set and the executable test plan. Coverage exclusions were limited to reviewed fixed-configuration or unreachable structural items; SVA is tracked separately from the functional coverage document.

Coverage metric	Merged result	Closure interpretation
URG score	96.76%	Final score after reviewed condition/toggle exclusions.
Line coverage	98.61%	Reachable RTL control and datapath statements exercised.
Condition coverage	98.19%	Remaining conditions are reviewed defensive or unreachable combinations.
Toggle coverage	90.19%	Reviewed fixed/reserved bits and parameter-disabled signals excluded.
FSM coverage	100.00%	All reachable manager and DMA state transitions exercised.
Branch coverage	90.31%	Remaining branches are reviewed structural/defensive paths.
Assertion coverage	100.00%	SVA properties activated and checked in targeted stall tests.
Functional group coverage	100.00%	All planned functional groups closed.

5. Scoreboard and Reference Model

The reference model is a cycle-independent byte-addressable memory model. Before each descriptor is submitted, the source region is initialized with expected data and the destination region is marked empty. AXI write observations update the model only on valid write handshakes and byte strobes. At completion, the scoreboard compares destination bytes against the expected source bytes and checks TAG, length, error status, and completion ordering.

The scoreboard is intentionally independent of the RTL implementation: it does not reuse the DMA burst splitter, FIFO pointers, or manager state machine. This prevents a duplicated RTL bug from being accepted as the reference result.

6. Assertions and Protocol Checks

Assertion family	Checked behavior	Activation scenario
AXI-Lite stability	AW/W/AR/R/B payload remains stable while VALID is held and READY is low.	SVA activation and AXI-Lite access tests.
Descriptor stability	Read/write descriptor fields remain stable during descriptor backpressure.	Descriptor stall and SVA activation.
AXI channel stalls	AR/R/AW/W/B valid-ready stalls are observed without data corruption or deadlock.	AXI backpressure and outstanding tests.
AXIS FIFO backpressure	Data and sidebands remain stable while downstream TREADY is low.	AXIS backpressure test.
Reset recovery	State, FIFO, and IRQ return to reset values and a subsequent descriptor completes.	Reset recovery test.

7. Coverage Review and Known Boundaries

The verified scope covers aligned transfers, AXI bursts, 4KB boundary splitting, read/write error responses, FIFO backpressure, multiple in-flight bursts, completion/IRQ behavior, reset recovery, RAL access, and descriptor queue management.

The following capabilities are intentionally outside the current positive feature scope: ENABLE_UNALIGNED=1 data shifting, scatter-gather linked-list descriptor fetch, multiple independent DMA engines/channels, out-of-order completion across descriptors, and fully implemented write_abort semantics.

8. Final Conclusion

The AXI4 DMA subsystem has completed the planned functional verification scope. The 28-test UVM regression exercises the control plane, descriptor admission, queueing, data movement, AXI protocol stress, error handling, completion/IRQ behavior, and reset recovery. The merged coverage database and reviewed exclusions provide a traceable basis for project sign-off.

Appendix A. Final Coverage Dashboard

Merged final result after reviewed exclusions:

Metric	Result
URG Score	96.76%
Line	98.61%
Condition	98.19%
Toggle	90.19%
FSM	100.00%
Branch	90.31%
Assertion	100.00%
Functional Group	100.00%